What Drives GitHub Repository Success? A Data-Driven Analysis of Open-Source Projects

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***Abstract*—GitHub is one of the largest platforms for hosting and sharing open-source software projects, yet the factors defining success of a repository remain an active area of research. This study investigates the characteristics influencing to GitHub repository popularity using a dataset containing the top 500 repositories ranked by stars.**

The results indicate that repository popularity is highly unevenly distributed, with a small number of repositories attracting a unevenly large share of community attention. A moderately strong positive correlation was observed between stars and forks (r = 0.608), suggesting that community engagement is closely associated with repository success. In contrast, repository size exhibited no relationship with popularity (r = -0.012). Additionally, Python, TypeScript, and JavaScript were found to be the most frequently used programming languages among highly popular repositories.

These findings suggest that community participation and engagement are more strongly associated with repository success rather than repository size or programming language. These results contribute to a better understanding of the factors related to popularity within open-source software ecosystems.

I. Introduction

GitHub is one of the most widely used platforms for hosting and sharing open-source software projects, used by millions of programmers and developers worldwide. However, the success of repositories on GitHub varies significantly, and not all projects achieve the same level of visibility or community engagement. Repository popularity is commonly measured using indicators such as stars and forks, which reflect user interest and external contributions.

Understanding the factors that defines repository success is important for both developers and researchers, as it can provide insights into what drives visibility and engagement in open-source ecosystems. This study investigates the key determinants of GitHub repository popularity. Specifically, this research addresses the following questions:

- What factors are most associated with GitHub repository popularity (measured by stars)?
- Which programming languages are most commonly represented among highly popular repositories?
- What is the relationship between repository popularity and characteristics such as forks and repository size?
- What distinguishes top 10% repositories from the rest?

II. Related Work

Defining the success of an open-source repository is not straightforward, as success can be defined and measured using different criteria. While some researchers evaluate success through popularity indicators such as stars and forks, others consider factors such as contributor activity, community engagement, project growth, and software adoption. Therefore, different approaches have been proposed to assess and analyze repository success on GitHub.

Borges et al. [1] proposed a framework for evaluating the popularity of GitHub repositories using the number of stars as the primary indicator. In addition, they introduced popularity growth patterns to describe how repository popularity changes over time. Their findings showed that stars are positively associated with other indicators of repository success. This work proved stars as a suggestive measure of repository popularity.

Al-Rubaye and Sukthankar [2] examined popularity on GitHub from a social-media perspective. They propsed that actions such as starring, watching, and forking repositories represent different forms of user engagement similar to interactions on social media platforms. Based on these indicators, they proposed a Weight-Based Popularity Score (WTPS), which combines multiple engagement metrics into a single measure of repository popularity.

Han et al. [3] investigated the characterization and prediction of popular GitHub projects. Using a large-scale dataset and machine learning techniques, they found factors that contribute to repository popularity. Their results showed that the number of branches, contributors, and open issues are among the most significant characteristics associated with popular repositories. Furthermore, their predictive model demonstrated that repository popularity can be estimated using project-related and community-related features.

While previous studies have primarily focused on popularity assessment frameworks, popularity scoring methods, and predictive models using large-scale datasets, this study uses an exploratory data analysis approach. Using a dataset containing the top 500 GitHub repositories ranked by stars [4], this work investigates the relationships between repository popularity, programming language distribution, repository size, and forking activity.

III. Dataset and Methodology

The dataset used for analysis in this study was obtained from Kaggle[4] and consists of the top 500 GitHub repositories ranked by the number of stars. The dataset contains information about GitHub repositories, including their names, descriptions, programming languages, star counts, fork counts, and other relevant metadata. During analysis, repositories with incomplete metadata were dropped to ensure reliability of the data.

The analysis was carried out using Python and the Pandas, NumPy, Matplotlib, and Seaborn libraries. First, an exploratory data analysis was performed to examine the distribution of repository popularity and the characteristics of the repositories in the dataset. Histograms and bar charts were used to visualize the distributions of stars and programming languages. All analyses were conducted in a Jupyter Notebook environment. The complete source code and analysis workflow are available in the project's GitHub repository [5].

To investigate relationships between repository characteristics, scatter plots were generated for stars versus forks and stars versus repository size. Additionally, Pearson correlation coefficients were calculated to define the strength of the relationships between these variables. The results were then interpreted to identify factors associated with repository popularity and to evaluate which different repository characteristics relate to GitHub success. Repositories in the top 10% by star count were additionally compared with the remaining repositories to find characteristics associated with highly popular projects.

IV. Results

A. Distribution of Repository Popularity

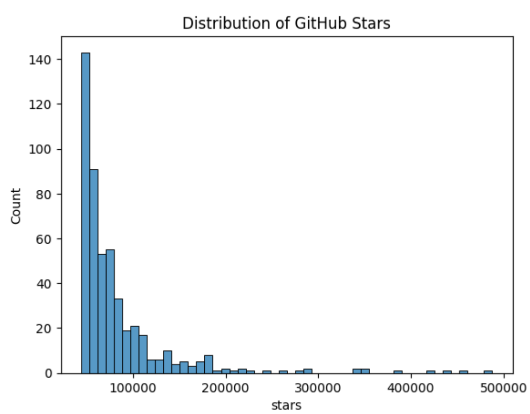

Fig. 1. Distribution of GitHub repository stars.

As shown in Figure 1, repository shows a highly skewed distribution, indicating that repository popularity is unevenly distributed. While most repositories accumulate between approximately 50,000 and 100,000 stars, only a small number achieve extremely high popularity, exceeding 200,000 stars. This suggests that a limited number of projects attract a unevenly large share of community attention.

B. Programming Language Distribution

As shown in Figure 2, Python is the most frequently represented programming language among the repositories in the dataset, followed by TypeScript and JavaScript. Together, these languages account for a major proportion of the top 500

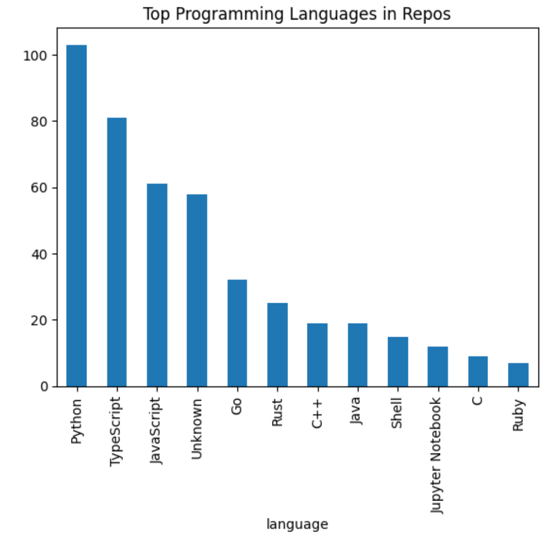

Fig. 2. Distribution of programming languages among the top 500 GitHub repositories.

repositories. This finding suggests that projects developed in these languages are strongly used among the most popular GitHub repositories. However, the distribution reflects the composition of the dataset and does not by itself imply that repositories written in these languages are more successful than those written in other languages.

C. Relationship between Stars and Forks

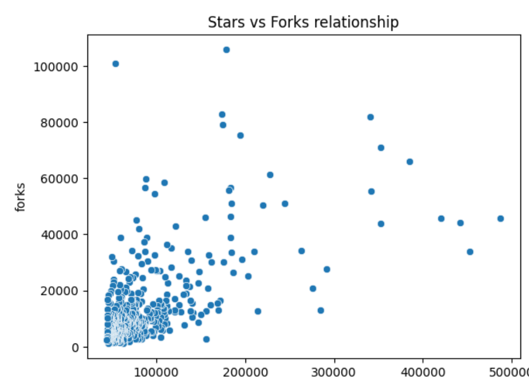

Fig. 3. Scatter plot of repository stars and forks.

As shown in Figure 3, a positive relationship can be observed between the number of stars and forks. Repositories with a higher number of stars generally tend to have more forks, suggesting that popular projects are also more likely to attract community engagement.

To identify this relationship, the Pearson correlation coefficient between stars and forks was calculated as $r = 0.608$. This value indicates a moderately strong positive correlation, showing that repositories with greater popularity tend to gather more forks. However, the relationship is not perfectly linear, as repositories with similar star counts may have different numbers of forks. This suggests that additional factors beyond popularity may influence community participation and repository forking behavior.

Overall, the results indicate that stars and forks are closely related characteristics and together provide valuable indicators of repository success within the GitHub ecosystem.

D. Repository Size and Popularity

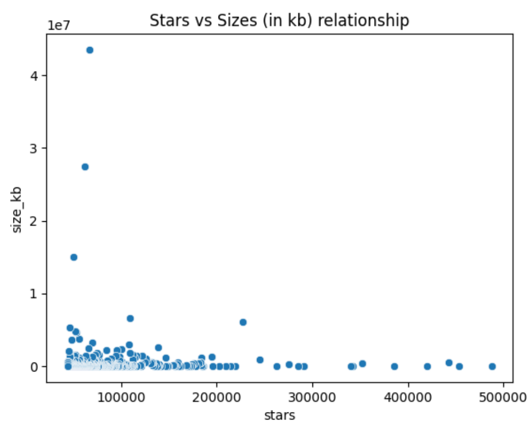

Fig. 4. Scatter plot of repository stars and repository size.

As shown in Figure 4, no clear relationship can be observed between repository size and popularity. The scatter plot reveals nearly zero correlation of repository sizes across different star counts, with highly starred repositories varying substantially in size.

To identify this relationship, the Pearson correlation coefficient between stars and repository size was calculated as $r = -0.012$. This value indicates an extremely weak negative correlation, effectively suggesting the absence of a linear relationship between the two variables. In other words, larger repositories are not necessarily more popular, and smaller repositories can achieve levels of popularity comparable to those of much larger projects.

These findings imply that repository size alone is not a meaningful predictor of success on GitHub. Other factors, such as project usefulness, community engagement, documentation quality, or development activity, are likely to play a more significant role in determining repository popularity.

E. Characteristics of Top 10% Repositories

TABLE I
COMPARISON OF TOP 10% REPOSITORIES AND THE REMAINING 90%

Metric	Top 10%	Remaining 90%
Average Stars	223,065	67,916
Average Forks	37,491	10,852
Average Size (in KB)	373,833	482,933

Table I summarizes the characteristics of repositories whose star counts exceed the 90th percentile and compares them with the remaining repositories in the dataset. As expected, repositories in the top 10% exhibit substantially higher average star counts. More importantly, they also show significantly higher levels of community engagement, with an average of 37,491 forks compared to 10,852 forks for the remaining repositories.

In contrast, repository size does not appear to distinguish highly popular repositories from less popular ones. The average repository size of the top 10% repositories is slightly smaller than that of the remaining repositories. This observation is consistent with the previously reported correlation analysis, which indicated nearly no relationship between repository size and popularity.

The programming language distribution among the top 10% repositories is dominated by Python, followed by TypeScript and JavaScript. Repositories with missing language metadata were excluded from this interpretation. These languages are also highly represented in the overall dataset, suggesting that they are common among highly starred GitHub projects.

Overall, the results indicate that community engagement, reflected by the number of forks, is more strongly associated with repository success than repository size or programming language. Highly popular repositories tend to attract substantially more developer participation while not necessarily being larger projects.

V. Discussion

The results of this study show several important characteristics of repository popularity on GitHub. First, the distribution of stars is highly skewed, implying that popularity is concentrated among a relatively small number of repositories. While many repositories achieve moderate levels of popularity, only a limited number attract exceptionally large communities and receive very high numbers of stars.

The analysis also demonstrated a moderately strong positive correlation between stars and forks $r = 0.608$. This finding suggests that repository popularity and community engagement are closely related. Repositories that attract more attention from users are also more likely to be forked and reused by developers. This observation is consistent with the findings of Borges et al. [1], who reported that stars are associated with forks and the effective use of software by third-party applications. Similarly, Al-Rubaye and Sukthankar [2] emphasized the importance of user interactions such as starring, watching, and forking as indicators of popularity and engagement on GitHub. Therefore, the results of the present study provide additional evidence that community participation plays a crucial role in repository success.

In contrast, repository size shoed nearly no relationship with popularity $r = -0.012$. This suggests that larger repositories are not necessarily more successful than smaller ones. A repository may contain a relatively small codebase while solving an important problem and attracting a large user community. Conversely, a large repository may have limited adoption despite its size. This finding highlights that repository usefulness and community interest are likely to be more important characteristics of success rather than the amount of source code alone.

Python was the most common programming language among the top repositories, followed by TypeScript and

JavaScript. This may reflect the growing importance of fields such as artificial intelligence, data science, and web development. However, since these languages are also highly represented in the overall dataset, the results do not imply a direct relationship between programming language and repository success.

The findings of this study generally support previous research. The positive relationship between stars and forks agrees with the observations of Borges et al. [1], while the importance of community engagement aligns with the work of Al-Rubaye and Sukthankar [2]. Additionally, the finding that repository success is influenced by multiple factors is consistent with the conclusions of Han et al. [3], who identified several repository and community characteristics associated with popularity.

Overall, the results suggest that repository success cannot be fully explained by a single metric. While stars provide a useful measure of popularity, forks offer an additional dimension on community engagement and developer participation. Together, these metrics provide a more comprehensive understanding of repository success on GitHub. However, the findings should be interpreted in light of the limitations of the study.

A. Limitations

This study has several limitations. First, the analysis was conducted using a dataset containing only the top 500 GitHub repositories ranked by stars. Previous studies have often relied on significantly larger datasets, allowing for more detailed analyses. Second, the study focused mainly on exploratory data analysis and considered only a limited number of repository characteristics. Factors such as contributor activity, issue management, commit history, documentation quality, and repository age were not examined and may also influence repository popularity.

B. Future Work

Future research could extend this work by using larger and more diverse datasets that include repositories with a wider range of popularity levels. Additional repository characteristics, such as contributor counts, commit activity, issue statistics, and project age, could also be expressed into the analysis. Also, machine learning techniques similar to those proposed by Han et al. [3] could be applied to develop predictive models of repository popularity and to identify the most significant factors contributing to repository success.

VI. Conclusion

This study investigated the factors associated with GitHub repository success using a dataset containing the top 500 repositories ranked by stars. Through exploratory data analysis, the relationships between repository popularity, programming language, repository size, and forking activity were examined.

The results showed that repository popularity is highly unevenly distributed, with a relatively small number of repositories attracting a unevenly large share of community attention. A moderately strong positive correlation was observed between stars and forks, indicating that popular repositories are more likely to generate greater community engagement and participation. In contrast, repository size showed nearly no relationship with popularity, suggesting that larger repositories are not necessarily more successful than smaller ones. The analysis also showed that Python, TypeScript, and JavaScript are strongly used among highly popular repositories.

Furthermore, the comparison between the top 10% of repositories and the remaining repositories demonstrated that highly successful projects tend to attract substantially more forks while not necessarily being larger in size. These findings suggest that community engagement is more closely associated with repository success than repository size.

Overall, the study highlights that GitHub repository success cannot be explained by a single factor. While popularity is influenced by multiple characteristics, indicators of community involvement, such as forks, appear to be among the strongest signals associated with successful open-source projects.

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